

Element K

Defining The Problem:

Element A- In element A we came up with our problem statement. Which was our criteria for the project and who this project will affect and just the overall purpose of the project. We also came up with the problem definition which was describing our project and explaining what this project truly is. Finally we went over the stakeholder concerns and connections which was just mainly who our clients were and who this project affected and the different connections between these clients.

Element B- In element B we looked at different plant platform designs to get an idea of how we wanted to design our project. We looked at 5 different types of plant platforms and described the pros and cons of each one. Each of these different designs helped us come to a conclusion on what we wanted our final design to look like.

Element C- In this element we each made a design for the project and compared each other's designs. We compared the looks of each project and showed how well each of our designs did, like how programable it is or how good its shade is.

What went well and what didn't go well with this project step: I would say that our understanding of the problem and research was very good and our construction of the different parts. I would also say that we could have added more to our project definition and found more connections between our stakeholders which

would have helped us when it came to who we were making our project for exactly.

Ideation:

Element D- In element D each of the group members came up with a design for our project. We graded each design on: sun protection, size, support, durability, programmability, waterproof, and decoration. We ended up landing on Ana's design. Ana's design was a rotating platform that would spin around once the plant had been in the sun for around 6 hours. Once it was in the shade for a while it would spin back into the sun. This was our final design for our plant platform.

What went well and what didn't go well with this project step: In this step we had very good designs and different perspectives on the project which helped us come up with more ideas. Something that didn't go well with this step was we found it a bit hard to come up with ideas which later caused some issues with understanding what to add to our project.

Selection & prototype design:

Some aspects of element d were used in this step. This was just mainly the stem and programming part of our process.

What went well and what didn't go well with this project step: In this step we did a good job with using a lot of different materials and topics to help us make our prototype. Something that didn't go so well was the durability of our project and the other concepts of the weather and animals which we didn't really think about.

Prototype building: We did of course build our project but we really didn't do an element g or put it into a document.

Testing:

Element H- In element H we listed each of our test predictions for our requirements in element D (sun protection, size, support, durability, programmability, waterproof, and decoration). We listed our testing methods and procedures, the test case and number of trials, the expected range of results, the criteria if it passed or failed, and goals set beyond criteria. We set this up to make sure everything went how we wanted it to go. We also listed our feedback from when the duke engineer came into the classroom.

What went well and what didn't go well with this project step: We were able to set up each testing method we wanted to do and we got feedback to fix our project, and make it better. Nothing really went wrong in this element.

Element I- In element I we focused on organizing and structuring all of our tests, data collection, and analysis. We followed the list of our main priorities and numbered them from highest to lowest. We started with sun protection as our highest priority and we talked about all the different tests we did for it and how our results came out to be. We then did the same going down our list which included size, support, durability, programmability, waterproof, and decoration.

What went well and what didn't go well with this project step: In this step we did a good job at setting up goals and estimating how we would set up our tests. Something that didn't go so well was our amount of tests on some priorities and less on others. It was very inconsistent and we could have maybe done more for the other priorities.

Evaluation:

Element J- In Element J we had to list the things about our project that we needed to improve based on feedback. We put durability and consistency which kind of go hand and hand. We put these because we were having to constantly fix it before the presentation. We said that the reason for these two problems were time and materials.

K3- During this project we had many ups and downs, lots of things didn't work and many times we had to reconsider/regroup. Our most difficult obstacle was mostly just actually getting it to spin. It was difficult making the track for the wheels to roll on and the part where we had to connect the motor to the spinner to the platform.